

Java Exception Handling: ArrayIndexOutOfBoundsException

Program: ArrayIndexDemo

1. What is ArrayIndexOutOfBoundsException?

ArrayIndexOutOfBoundsException is an unchecked runtime exception in Java. It occurs when a program tries to access an array using an index that is outside the valid range. For an array of length 3, the valid indexes are 0, 1, and 2. Trying to access index 5 causes this exception.

Example: In this program, numbers contains three elements, so numbers[5] is invalid.

2. Objective

To understand and demonstrate how Java handles an ArrayIndexOutOfBoundsException using the try-catch mechanism when an invalid array index is accessed.

3. Task Description

Create a Java program that stores values in an integer array and deliberately attempts to access an invalid index. Catch the resulting ArrayIndexOutOfBoundsException and display a meaningful message instead of allowing the program to terminate abnormally.

4. Step-wise Execution

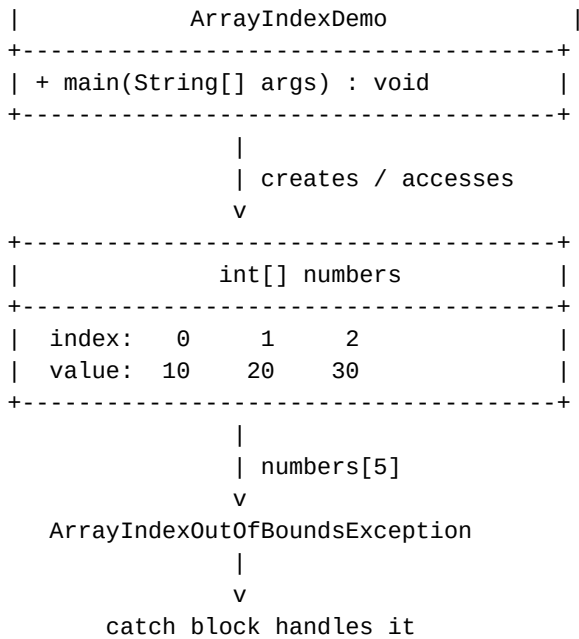
1. The program starts execution from the main() method.
2. An integer array numbers is created with values 10, 20, and 30.
3. The array length is 3, so valid indexes are 0, 1, and 2.
4. The statement numbers[5] attempts to access index 5.
5. Java detects that index 5 is outside the valid range and throws ArrayIndexOutOfBoundsException.
6. The catch block catches the exception.
7. The message Exception: Array index is out of bounds is printed.

5. Algorithm

1. Start.
2. Create an integer array with three elements.
3. Enter the try block.
4. Access an invalid array index.
5. If ArrayIndexOutOfBoundsException occurs, transfer control to the catch block.
6. Display an appropriate exception message.
7. Stop.

6. UML Diagram

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7. Program Code

```

class ArrayIndexDemo {
    public static void main(String[] args) {
        try {
            int[] numbers = {10, 20, 30};
            System.out.println(numbers[5]);
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Exception: Array index is out of bounds");
        }
    }
}

```

8. Code Explanation

`class ArrayIndexDemo` — Defines the class containing the program.

`public static void main(String[] args)` — The main method is the starting point of execution.

`try` — Contains code that may generate an exception.

`int[] numbers = {10, 20, 30};` — Creates an integer array of length 3.

`System.out.println(numbers[5]);` — Attempts to access index 5. This is invalid because only indexes 0-2 exist.

`catch (ArrayIndexOutOfBoundsException e)` — Catches the exception generated by the invalid index.

`System.out.println(...)` — Prints a user-friendly error message.

9. Expected Output / Output Screenshot

When the program is executed, the output is:

Output

Exception: Array index is out of bounds

Program completed without crashing.

Result: The program successfully demonstrates how `ArrayIndexOutOfBoundsException` is generated and handled using try-catch.